

Perfluorinated Compounds, Polychlorinated Biphenyls, and Organochlorine Pesticide Contamination in Composite Food Samples from Dallas, Texas, USA

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Objectives The objective of this article is to extend our previous studies of persistent organic pollutant (POP) contamination of U.S. food by measuring perfluorinated compounds (PFCs), organochlorine pesticides, and polychlorinated biphenyls (PCBs) in composite food samples. This study is part of a larger study reported in two articles, the other of which reports levels of polybrominated diphenyl ethers and hexabromocyclododecane brominated flame retardants in these composite foods [Schechter et al. 2010. Polybrominated diphenyl ethers (PBDEs) and hexabromocyclododecane (HBCD) in composite U.S. food samples, *Environ Health Perspect* 118:357–362]. **Methods** In this study we measured concentrations of 32 organochlorine pesticides, 7 PCBs, and 11 PFCs in composite samples of 31 different types of food (310 individual food samples) purchased from supermarkets in Dallas, Texas (USA), in 2009. Dietary intake of these chemicals was calculated for an average American. **Results** Contamination varied greatly among chemical and food types. The highest level of pesticide contamination was from the dichlorodiphenyltrichloroethane (DDT) metabolite p,p'-dichlorodiphenyldichloroethylene, which ranged from 0.028 ng/g wet weight (ww) in whole milk yogurt to 2.3 ng/g ww in catfish fillets. We found PCB congeners (28, 52, 101, 118, 138, 153, and 180) primarily in fish, with highest levels in salmon (PCB-153, 1.2 ng/g ww; PCB-138, 0.93 ng/g ww). For PFCs, we detected perfluorooctanoic acid (PFOA) in 17 of 31 samples, ranging from 0.07 ng/g in potatoes to 1.80 ng/g in olive oil. In terms of dietary intake, DDT and DDT metabolites, endosulfans, aldrin, PCBs, and PFOA were consumed at the highest levels. **Conclusion** Despite product bans, we found POPs in U.S. food, and mixtures of these chemicals are consumed by the American public at varying levels. This suggests the need to expand testing of food for chemical contaminants.